

Human plasma derived exosomes: Impact of active and passive drug loading approaches on drug delivery

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Abstract

The aim of the current study was to explore the potential of human plasma-derived exosomes as versatile carriers for drug delivery by employing various active and passive loading methods. Exosomes were isolated from human plasma using differential centrifugation and ultrafiltration method. Drug loading was achieved by employing sonication and freeze thaw methods, facilitating effective drug encapsulation within exosomes for delivery. Each approach was examined for its effectiveness, loading efficiency and ability to preserve membrane stability. Methotrexate (MTX), a weak acid model drug was loaded at a concentration of 2.2 μM to exosomes underwent characterization using various techniques such as particle size analysis, transmission electron microscopy and drug loading capacity. Human plasma derived exosomes showed a mean size of 162.15 ± 28.21 nm and zeta potential of -30.6 ± 0.71 mV. These exosomes were successfully loaded with MTX demonstrated a better drug encapsulation of 64.538 ± 1.54 % by freeze thaw method in comparison 55.515 ± 1.907 % by sonication. In-vitro drug release displayed 60 % loaded drug released within 72 h by freeze thaw method that was significantly different from that by sonication method i.e., 99 % within 72 h (p value 0.0045). Moreover, cell viability of exosomes loaded by freeze thaw method was significantly higher than that by sonication method (p value 0.0091) suggested that there was membrane disruption by sonication method. In conclusion, this study offers valuable insights into the potential of human plasma-derived exosomes loaded by freeze thaw method suggest as a promising carrier for improved drug loading and maintenance of exosomal membrane integrity.